Topic 24 — Current Environmental Issues

Covers syllabus (click to expand)

Microplastics | Carbon Credit (कार्बन क्रेडिट) | Carbon Trading | Net Zero Emissions (नेट जीरो उत्सर्जन) | Green Finance (हरित वित्त) | Blue Economy (नीली अर्थव्यवस्था) | LiFE Mission

Sources baked in: UNEP; MoEFCC; SEBI; RBI; IPCC; UN Plastics Treaty; Green Credit Rules 2023; CCTS 2023; UPPCS PYQs 2018–2025 **Weight:** ★★★ High — LiFE COP trap; carbon credit vs trading; India net zero 2070; microplastic size **Last verified:** August 2026

Current Affairs (this topic)

- This chapter is the CA layer — keep NbS, blue economy, and plastics stems current.
- UP is landlocked: do not invent statewide blue-economy projects.

24.1 Microplastics

Identity: <5 mm | Primary vs secondary | Persistent | Ganga (गंगा)/food-chain contamination

- **Microplastics** are plastic particles **smaller than 5 mm** — primary (manufactured small) or secondary (fragmented from larger plastic).
- **Primary:** microbeads in cosmetics, plastic **nurdles**, industrial abrasives.
- **Secondary:** bags, bottles, fishing nets, and **tyre wear** breaking down — most environmental mass is secondary.
- **Major sources:** single-use packaging, **synthetic textile fibres** (laundry effluent), tyre abrasion, fishing gear, agricultural plastic mulch.
- **Pathways:** land → rivers → sea. **Ganga and Yamuna (यमुना)** studies show microplastic contamination downstream of urban (नगरीय) centres (Kanpur (कानपुर), Varanasi (वाराणसी), Lucknow (लखनऊ)).
- **Health concern:** detected in blood, lungs, placenta — potential carrier for toxins; research ongoing.
- **Persistence trap:** polyethylene/polypropylene persist **decades to centuries** — photodegradation makes **smaller** particles, not harmless minerals.
- **Global policy:** UNEA **2022** resolution launched **Global Plastics Treaty** negotiations (INC process).
- **India:** Plastic Waste Management Rules (EPR); **ban on identified single-use plastic items (July 2022)**; National Dashboard for EPR.

Type	Origin	Example
Primary	Manufactured small	Microbeads, nurdles
Secondary	Large plastic breaks down	Bag fragments, tyre particles

> **Logic:** Microplastics = **<5 mm**, not 5 cm. "Biodegrade within weeks" = **FALSE**.

24.2 Carbon Credit

Identity: 1 credit = 1 tCO_2e | Kyoto CDM/CERs | Nature-based credits | Green Credit ≠ carbon credit

- **Carbon credit** is a tradable certificate representing **one tonne of CO_2 equivalent (tCO_2e)** reduced, avoided, or removed.
- **Project-based generation:** verified emission-reduction project (solar displacing coal, biogas (बायोगैस), afforestation) → accredited body issues credits → sold to buyer.
- **Kyoto CDM (Clean Development Mechanism):** developed countries fund projects in developing countries → **Certified Emission Reductions (CERs)**. India hosted many wind/solar CDM projects.
- **Voluntary Carbon Market:** companies buy credits for net-zero pledges — **Verra (VCS), Gold Standard** standards.
- **Nature-based credits:** afforestation, REDD+, **mangrove restoration** — permanence risk if forest burns or is cleared.
- **Avoidance vs removal:** avoidance prevents emissions that would occur; removal pulls CO_2 from atmosphere (afforestation, DAC).
- **Additionality:** project must prove reduction would not happen without credit revenue.
- **Green Credit Programme (2023)** rewards tree plantation, water conservation, waste management — **NOT the same as carbon credit** (broader eco-action incentive, not strictly 1 tCO_2e unit).

PYQ — UPPCS Prelims 2021, Q125

The concept of 'carbon credit' originated from:

A. Earth Summit, Rio

B. Kyoto Protocol

C. Montreal Protocol (मॉन्ट्रियल प्रोटोकॉल)

D. G-8 Summit

Show answer

Ans: B — Kyoto Protocol's **CDM and flexible mechanisms** established carbon credit architecture.

> **Logic:** Green Credit Programme **2023 ≠ carbon credit**. Double counting avoided under Paris Article 6.

24.3 Carbon Trading

Identity: Cap-and-trade | EU ETS largest | Kyoto origin | India CCTS 2023 | PAT = ESCerts

- **Carbon trading** is the **market mechanism** to buy and sell emission **allowances or credits** under a regulatory cap.
- **Cap-and-trade:** regulator sets total emissions cap → firms get allowances → efficient firms sell surplus to high emitters → cap met at lowest cost.
- **Carbon credit vs carbon trading:** credit = **unit** (1 tCO_2e); trading = **market** exchanging those units.
- **EU Emissions Trading System (EU ETS)** is the world's **largest compliance carbon market**.
- **Kyoto flexible mechanisms:** (1) Emissions Trading among Annex I countries, (2) **CDM**, (3) Joint Implementation.
- **India Carbon Credit Trading Scheme (CCTS), 2023** under **Energy Conservation Act 2001 — Bureau of Energy Efficiency (BEE)** (ऊर्जा दक्षता ब्यूरो) as designated authority.
- **PAT (Perform, Achieve and Trade (पण्याध्यक्ष)), 2012:** trades **energy efficiency certificates (ESCerts)** — parallel market-based compliance, **not identical to carbon credits**.
- **Compliance vs voluntary:** compliance markets are legally mandated (EU ETS, future CCTS); voluntary markets are optional corporate purchases.

PYQ — UPPCS Prelims 2024, Q35

Which international agreement is related to climate change (जलवायु परिवर्तन) mitigation?

A. Montreal Protocol

B. Ramsar (रामसर) Convention

C. Kyoto Protocol

D. CITES (साइट्स)

Show answer

Ans: C — Kyoto = GHG targets + flexible mechanisms. Montreal = **ozone**, not climate.

Concept	What it is
Carbon credit	Unit = 1 tCO_2e
Carbon trading	Market exchanging credits/allowances
Carbon offset (कार्बन ऑफसेट)	Using credits to compensate own emissions

> **Logic:** Montreal = ozone. Kyoto/Paris = climate. PAT = **energy efficiency**, not carbon directly.

24.4 Net Zero Emissions

Identity: Emissions = removals | India 2070 COP26 | NDC 2030 interim | Emissions Gap = UNEP

- **Net zero:** anthropogenic GHG emissions are **balanced by anthropogenic removals** over a specified period — not absolute (निरपेक्ष) gross zero.
- **Gross zero** means eliminating virtually all emissions — harder and costlier than net zero.
- **India's target: net zero by 2070** — announced by PM at **COP26 Glasgow, November 2021**.
- **Interim NDC 2030 (updated 2022):** (1) reduce emissions **intensity of GDP by 45%** from 2005, (2) **50% cumulative electric capacity from non-fossil sources**, (3) additional **2.5–3 $GtCO_2e$ carbon sink (कार्बन सिंक)** through forests.
- **Carbon neutral vs net zero:** carbon neutral often means **CO_2 only**; full GHG net zero includes methane and N_2O.
- **Country comparison:** EU/UK **2050**, China **2060**, India **2070**, US **2050** (pledge).
- **UNEP Emissions Gap Report** measures gap between NDC pledges and **1.5°C/2°C pathways** — published by **UNEP** (संयुक्त राष्ट्र पर्यावरण कार्यक्रम).

Country	Net zero year
EU / UK	2050
China	2060
India	**2070**

> **Logic:** India net zero = **2070**, not 2050. Net zero ≠ no fossil fuels from tomorrow.

24.5 Green Finance

Identity: Green bonds | Sovereign Green Bonds FY 2022–23 | SEBI framework | GCF | Loss and Damage COP28

- **Green finance** mobilises capital for investments with environmental benefits — clean energy, pollution prevention, biodiversity, climate adaptation (जलवायु अनुकूलन).
- **Green bonds:** debt where proceeds fund **eligible green projects** only — renewables, clean transport, water, adaptation.
- **India's first Sovereign Green Bonds** issued in **FY 2022–23** (January 2023 tranches) for solar, wind, green hydrogen, EV public transport.
- **SEBI Green Bond Framework (2023)** defines eligible categories and disclosure — aligns with ICMA Green Bond Principles.
- **Green Climate Fund (GCF)** (जीसीएफ) operates under **UNFCCC** (यूएनएफसीसीसी) — established at **Cancun COP-16 (2010)** to support developing-country adaptation and mitigation. It is **not** a Durban creation and not founded under UNEP/OECD/ADB/World Bank.
- **100 billion/year climate finance pledge** by developed countries — committed at **COP15 Copenhagen (2009)**, reaffirmed Paris 2015.
- **Loss and Damage Fund** operationalised at **COP28 UAE (2023)** — finances recovery from climate impacts exceeding adaptation capacity.

- **Green finance ≠ Green Credit Programme** — finance = capital markets; Green Credit = eco-action incentive certificates.

Instrument	India milestone
Sovereign Green Bonds	FY **2022–23**
SEBI Green Bond Framework	**2023**
Loss and Damage Fund	**COP28 2023**

> **Logic:** Sovereign Green Bonds = **FY 2022–23**. Loss and Damage Fund = **COP28**, not COP26.

24.6 Blue Economy

Identity: SDG 14 | Sustainable ocean use | MISHTI 2023 | Sagarmala | Blue carbon

- **Blue economy** is the **sustainable use of ocean resources** for economic growth and livelihoods while preserving marine and coastal ecosystem (पारिस्थितिकी तंत्र) health.
- **SDG 14** = **Life Below Water** — conserve and sustainably use oceans, seas, and marine resources.
- India has **~2.37 million sq km EEZ** — fisheries, ports, shipping, coastal tourism, offshore wind, marine biotechnology.
- **Blue carbon:** mangroves, seagrass, salt (लवणाध्यक्ष) marshes sequester carbon faster per hectare than many terrestrial forests.
- **MISHTI** (Mangrove Initiative for Shoreline Habitats and Tangible Income) announced in **Union Budget 2023–24** — restore mangrove forests for cyclone protection and livelihoods.
- **Sagarmala:** port-led coastal development — MoPSW nodal.
- **Deep Ocean Mission:** manned submersible (Samudrayaan), deep-sea mining exploration, biodiversity survey.
- **UP is landlocked** — blue economy questions are national (MISHTI, Sagarmala), not UP coastal schemes.

PYQ — UPPCS Prelims 2025, Q93

MISHTI: (1) It was initiated in 2019. (2) It aims to restore mangrove forests.

A. Only 2

B. Neither

C. Both

D. Only 1

Show answer

Ans: A — Restoration aim correct. MISHTI = **2023–24 Budget**, not 2019.

Sector	Scheme
Mangrove restoration	MISHTI 2023
Ports/shipping	Sagarmala
Deep-sea exploration	Deep Ocean Mission

> **Logic:** MISHTI = **coastal mangroves**, not Ganga basin (गंगा). Blue economy = **SDG 14**.

24.7 LiFE Mission

Identity: Lifestyle for Environment | MoEFCC | COP26 idea 2021 | Launched June 2022 | 75 actions

- **LiFE (Lifestyle for Environment)** is an India-led movement for **individual and community pro-planet behaviour change** — mindful consumption over destructive consumption.
- **Idea announced at COP26, Glasgow, November 2021** by the PM — complement to government climate policy; individuals as change agents for SDG 12 and SDG 13.
- **Mission LiFE formally launched in June 2022 — MoEFCC** (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय) is nodal ministry.
- **75 pro-planet actions:** save energy, reduce waste, save water, sustainable food, e-waste disposal, tree plantation — via **Mission LiFE app/portal**.
- **Mechanism:** nudge and awareness (not regulatory law) — reduces lifestyle carbon footprint from food, transport, household energy.
- Supports **net zero 2070** through demand-side emission reduction alongside technology deployment.

PYQ — UPPCS Prelims 2025, Q139

LiFE: (1) It was launched in June 2022. (2) The idea was proposed at COP-25 in Madrid, Spain.

A. Only 2

B. Neither

C. Both

D. Only 1

Show answer

Ans: D — Launch **June 2022** true. Idea at **COP26 Glasgow 2021**, not COP-25 Madrid 2019.

Field	Detail
Idea	COP26 Glasgow **2021**
Launch	**June 2022**
Nodal	**MoEFCC**

Field	Detail
Actions	**75** pro-planet behaviours

> **Logic:** LiFE = **COP26 2021 / launch 2022** — COP-25 Madrid trap is **FALSE**.
